

Cloud and DevOps Interview Questions

Linux

1. Explain the difference between hard links and soft links.
2. How do you check disk space usage in Linux?
3. What does the command 'chmod 755' mean?
4. Explain inode in Linux.
5. How do you troubleshoot high CPU usage in Linux?
6. Difference between 'cron' and 'systemd' timers.
7. How to find the top 10 largest files in a directory.
8. How to check open ports in Linux.
9. What are runlevels?
10. Difference between '/etc/passwd' and '/etc/shadow'.
11. Explain SELinux and its modes.
12. How to check failed login attempts.
13. How to check last reboot time.
14. Difference between 'ps -aux' and 'top'.
15. Explain what happens when you type a command and hit enter.
16. Difference between 'kill', 'kill -9', and 'pkill'.
17. How to check memory usage in Linux.
18. Explain sticky bit in Linux.
19. How to mount a filesystem.
20. How to extend a filesystem.

Terraform

1. What is Terraform and why is it used?
2. Difference between Terraform and ARM templates/CloudFormation.
3. Explain Terraform state file and why it is important.
4. How do you manage remote state?

5. What is a Terraform provider?
6. Difference between 'terraform apply' and 'terraform plan'.
7. Explain Terraform modules.
8. How do you handle secrets in Terraform?
9. How do you manage multiple environments in Terraform?
10. Explain 'terraform import'.
11. What is the purpose of 'terraform fmt' and 'terraform validate'?
12. Difference between 'local-exec' and 'remote-exec'.
13. How to lock Terraform state?
14. How to roll back infrastructure changes in Terraform?
15. Explain workspaces in Terraform.
16. How do you upgrade provider versions?
17. Explain lifecycle rules in Terraform.
18. What are data sources in Terraform?
19. How do you debug Terraform issues?
20. How to integrate Terraform with CI/CD pipelines?

Azure

1. Explain Azure Resource Groups.
2. Difference between Azure VM Scale Sets and Availability Sets.
3. How does Azure Load Balancer differ from Application Gateway?
4. What is Azure Active Directory?
5. How do you implement RBAC in Azure?
6. Explain Azure Storage account types.
7. How to secure secrets in Azure?
8. Explain Azure Key Vault.
9. Difference between Azure Functions and Logic Apps.
10. Explain Azure Monitor and Application Insights.
11. How to implement high availability in Azure.
12. Difference between Azure SQL Database and Managed Instance.
13. How to configure Azure Virtual Network peering.
14. What is Azure Bastion?
15. Explain NSG vs ASG.

16. Explain Azure Policy.
17. How to set up alerts in Azure Monitor.
18. How do you manage costs in Azure?
19. Explain ExpressRoute.
20. Difference between Public, Private, and Hybrid cloud in Azure.

Google Cloud (GCP)

1. Explain GCP projects and IAM.
2. Difference between GCE and GKE.
3. How to configure service accounts in GCP.
4. Explain Cloud Storage classes.
5. Difference between GCP Load Balancers.
6. Explain VPC and subnets in GCP.
7. How to implement firewall rules in GCP.
8. Explain GCP Cloud Functions.
9. Explain Cloud Spanner vs Cloud SQL.
10. How to enable Stackdriver Monitoring.
11. What is GCP Cloud Pub/Sub?
12. Explain GCP Deployment Manager vs Terraform.
13. How do you implement logging in GCP?
14. Explain GCP BigQuery.
15. Difference between GCP Standard and Preemptible VMs.
16. Explain Cloud Run.
17. How do you secure API keys in GCP?
18. How to configure inter-VPC connectivity.
19. Explain Shared VPC.
20. Explain GCP Identity-Aware Proxy (IAP).

Docker

1. What is Docker and why is it used?
2. Explain difference between Docker image and container.
3. Explain Dockerfile and its instructions.
4. What is Docker Compose?
5. Difference between CMD and ENTRYPOINT in Dockerfile.
6. How do you check running containers?
7. Explain Docker networking modes.
8. How do you persist data in Docker?
9. Difference between volumes and bind mounts.
10. How do you optimize Docker images?
11. How to reduce image size in Docker.
12. Explain multi-stage builds.
13. How to debug Docker container issues.
14. Explain Docker Swarm.
15. Difference between Docker and VM.
16. How to secure Docker containers.
17. Explain 'docker exec' vs 'docker run'.
18. How do you monitor Docker containers?
19. Explain 'docker inspect'.
20. How to limit CPU and memory usage in Docker.

Kubernetes

1. What is Kubernetes and why is it used?
2. Explain Kubernetes architecture.
3. Difference between ReplicaSet and Deployment.
4. Explain Kubernetes services (ClusterIP, NodePort, LoadBalancer).
5. What are ConfigMaps and Secrets?
6. Explain Kubernetes namespaces.
7. How do you troubleshoot a failing pod?
8. Explain StatefulSets vs Deployments.
9. How to implement HPA (Horizontal Pod Autoscaler).
10. Explain RBAC in Kubernetes.

11. Difference between Ingress and LoadBalancer.
12. Explain DaemonSets.
13. Explain Kubernetes etcd.
14. How to perform rolling updates and rollbacks.
15. Explain Kubernetes CRDs.
16. How do you secure Kubernetes cluster?
17. Difference between kube-proxy and CNI plugins.
18. Explain Kubernetes taints and tolerations.
19. How do you debug CrashLoopBackOff errors?
20. How do you configure Kubernetes monitoring?

CI/CD

1. What is CI/CD and why is it important?
2. Difference between GitHub Actions, Jenkins, and GitLab CI.
3. Explain a typical CI/CD pipeline.
4. How to implement code quality checks in CI/CD.
5. How do you integrate Terraform with CI/CD?
6. Explain blue-green vs canary deployment.
7. How to implement secrets in CI/CD pipelines.
8. Explain how to roll back deployments via CI/CD.
9. How to integrate SonarQube in CI/CD.
10. How do you handle approvals in CI/CD?
11. How to optimize CI/CD pipeline performance.
12. How to implement artifact management.
13. Explain GitOps and ArgoCD.
14. How do you implement automated testing in CI/CD?
15. How to secure CI/CD pipelines.
16. Explain pipeline stages in Jenkins.
17. How do you handle parallel execution?
18. Difference between declarative and scripted pipelines.
19. How to monitor CI/CD pipelines.
20. Explain pipeline as code.

Monitoring and Security

1. Explain difference between monitoring and logging.
2. What is Prometheus and how does it work?
3. Explain Grafana dashboards.
4. How to configure alerts in Prometheus.
5. Explain ELK stack.
6. Difference between SIEM and log analytics.
7. How do you secure Kubernetes cluster?
8. Explain role of Service Mesh (Istio/Linkerd).
9. How do you monitor containers?
10. Explain Cloud Security Best Practices.
11. What is vulnerability scanning?
12. Explain OWASP Top 10.
13. How to perform penetration testing in cloud.
14. Explain Zero Trust model.
15. How do you handle secrets in CI/CD.
16. Explain role-based access control.
17. How do you enable audit logs.
18. Explain CIS benchmarks.
19. How to handle DDoS protection in cloud.
20. Explain Cloud WAF.

Bonus (Git, Networking, SRE)

1. Difference between Git merge and rebase.
2. Explain Git branching strategies.
3. What is a Git detached HEAD state?
4. How to resolve merge conflicts in Git.
5. Explain TCP vs UDP.
6. Difference between HTTP/1.1, HTTP/2, and HTTP/3.
7. Explain DNS resolution process.
8. Difference between VPN and VPC.
9. Explain SRE principles.
10. What are SLIs, SLOs, and SLAs?

11. Explain error budgets.
12. How do you measure reliability?
13. Explain incident management.
14. How to conduct blameless postmortems.
15. Explain chaos engineering.
16. Difference between monitoring and observability.
17. What is distributed tracing?
18. Explain canary releases in SRE.
19. How do you reduce toil as an SRE?
20. Explain service discovery.